signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
t = t_min:dt:t_max; // Time vector
// Define Signal 1: Rectangular pulse of width 2 centered at t = 0
x_t = (t >= -1) .* (t <= 1);
// Define Signal 2: Exponential decay function
h_t = exp(-t) .* (t >= 0); // Exponential decay, causal
// Perform the convolution
y_t = conv(x_t, h_t) * dt; // Multiply by dt to scale correctly
// Adjust the time vector for the convolution result
t_conv = (2 * t_min):dt:(2 * t_max);
// Plot Signal 1
clf; // Clear the current figure
subplot(3,1,1);
plot(t, x_t, 'b');
xlabel('t');
ylabel('x(t)');
title('Signal 1: Rectangular Pulse');
// Plot Signal 2
subplot(3,1,2);
plot(t, h_t, 'g');
xlabel('t');
ylabel('h(t)');
title('Signal 2: Exponential Decay');
// Plot the Convolution Result
subplot(3,1,3);
plot(t_conv, y_t, 'r');
xlabel('t');
ylabel('y(t)');
title('Convolution of x(t) and h(t)');
```

